CAPSTONE PROJECT PROPOSAL

Title: **A Decentralized Freelance Protocol with Web3 Integration**

Date: 10/02/2026

February - 2026

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Date: 2026/02/10

General Information

Project Title: Decentralized Freelance Protocol with Web3 Integration

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Executive Summary

Most of the online freelancing platforms are centralized which tends to create issues of high platform charges, slow payment, poor identity checks, lack of clarity in project processes and poor trust between clients and freelancers. Freelancers also cannot control their identities, data and work history which means that they rely on the platform and can be exposed to data misuse or system collapse.

In order to solve these problems, this project suggests designing and developing end-to-end web-based freelancing platforms based on a hybrid decentralized approach. The system is designed to enhance trust, transparency and collaboration by integrating decentralized technology with manageable web based solutions.

The proposed platform is based on identity management using a wallet-based system with help of MetaMask enabling users to be authenticated safely without the use of standard usernames and passwords. Project agreement, milestones and verifying transactions are managed in an open and impeccable manner by smart contracts created in Solidity. Decentralized storage (IPFS via Kubo) is used to store project documents and delivered to minimize the use of the centralized servers. Besides that a matching system that operates on rules and is developed with FastAPI suggests appropriate freelancers according to their skills, roles, availability and the nature of the project whereas PostgreSQL handles structured data on the platform.

This project is expected to deliver a working web application prototype that proves the existence of decentralized identity, secure project processes, project matching by rule, and trustworthy data storage over a well-defined academic scope.

1. Background and Literature Review

1.1 Background

Rapid digitalization and transformation of the global labor market from a traditional working model to a crowdsourcing/ freelancing model has enabled companies and professionals to access a global talent pool, providing operational scalability to clients **(Bari et al., 2015)**. However, it has also introduced issues such as security risks, worker well-being, and disputes between clients and freelancers **(Alvarez de la Vega et al., 2023; Shahzad, 2024)**. Current freelancing applications operate through a centralized architecture which results in high intermediary fees, often ranging from 5% to 20% **(Singh et al., 2024; Tamsetwar et al., 2026)**. Furthermore, the opaque dispute resolution process, reputation scoring, creates a "lock-in" effect where freelancers are not owners of their professional data **(Gandini et al., 2016)**.

Currently, centralized systems face issues such as a single point of failure, data breach, lack of financial transparency, while decentralized systems offer solutions to these issues, they often struggle with user adoption **(Tamsetwar et al., 2026)**. To address these gaps, this project intends to utilize a hybrid decentralized model, implementing a decentralized storage system, and Distributed Ledger Technology (DLT), while maintaining high-fidelity User Experience (UX) to resolve the inefficiencies in the current gig economy.

1.2 Literature Review Process

A systematic literature search was conducted primarily through Google Scholar, IEEE Xplore, ACM Digital Library and other subsequent databases to evaluate the socio-technical and architectural implications of the gig platforms. To maintain relevance and capture the transition of freelancing from early crowdsourcing to decentralized models, the search was restricted to the last decade.

An initial pool of 40 research papers were identified, which were then filtered through relevance to research, keywords such as freelancing, gig economy, blockchain, distributed security responsibility, IT workforce in the gig economy, digital labor platforms, Ethereum-based decentralized platforms and Mobile Crowdsourcing. Furthermore, identified literature was subjected to a screening process based on thematic relevance, where sources were excluded if they included non-digital gig sectors, such as ride sharing or delivery services, or localized country-specific studies to remain applicable to global architecture, and methodology used in research. This filtering ensured we remained focused on the technical and global aspect of freelancing.

Ultimately, 15 research papers were chosen for the literature review. The studies chosen were on the basis of socio-technical challenges of the current gig economy, and evaluating the technical feasibility of decentralized solutions.

1.3 Literature Review

1. Socio Technical Hurdles and Reputation Economy

In the early stages of crowdsourcing, researchers focused on the feasibility and repeatability of freelancing and gig economy as a whole; where lessons from major platforms indicated several issues such as "human uncertainty" are primary hurdles for the platforms growth in achieving reliability **(Bari et al., 2015)**. The author further identifies several key areas where freelancing excels, some of which include when projects are deemed impractical, when projects require niche skills, when projects require flexible time unfit for centralized staff. Multiple case studies such as eBay marketplace Globalization or PayPal Global Validation validated this through enabling the creation of scrum teams **(Bari et al., 2015)**.

To address leading issues relating to certainty within these platforms, "reputation economy" was established which acted as a proxy for trust within the environment **(Gandini et al., 2016)**. In a study of 9,593 freelancers on Elance, Gandini discovered that earnings were significantly correlated with reputation scores, however, conversely there existed no correlation between repeat jobs and reputation scores. Meaning, while reputation scores acted as a trust metric, it created significant "identity ambiguity" for new entrants who needed to navigate through "self-directed socialization" **(Blaising & Dabbish, 2022)**. Furthermore, qualitative studies highlight how freelancing can fragment work-life balance, requiring "worker-centric" designs **(Alvarez de Vega et al., 2023)**.

2. Functional Gaps: Collaboration vs. Security

As the sector matured, research shifted toward systematic management and worker experience, where considerable research was performed on development areas where freelancers are employed, and the issues associated with freelancer involvement in development processes **(Gupta et al., 2020)**. The specific distribution of these hurdles, as categorized by Guptaa et al. (2020), is detailed below:

Issues & Challenges	Percentage Reported (%)
Budget Estimation (Task Value Estimation)	8%
Capacity Utilization	3%
Collaboration & Coordination	33%
Communication	8%
Developer Recommendation (or selection)	19%
Intellectual Property Issues	6%
Participation of Crowd Worker	6%
Privacy and Security	11%
Recognition	8%
Task Decomposition	11%
Team Formulation	14%
Task Recommendation	14%
Trust Issues	8%
Market Dynamism	6%

Table 1: Issues and Challenges reported in Freelancing

The obtained data reveals that freelancers are less concerned regarding “Trust” within the system at 8% and more interested in “Collaboration” between each other being the primary hurdle at 33%. This also correlates with other studies, where freelancers are increasingly interested in cooperation, with 82% of respondents holding a positive view of collaborative work **(Fulker & Riedl, 2024)**. Furthermore, current platforms face other “weak points” such as privacy breaches, data leaks,etc **(Shahzad, 2024)**. Because freelancers work in resource lacking environments i.e.

their homes, **Rauf et al. (2023)** argues for a "distributed security responsibility" model, where security is shared between platform and developer.

3. Shift to Decentralized & Hybrid Solutions

Whilst addressing the security and trust concerns, studies prompted the use of decentralized platforms to mitigate issues such as payment, dependency on a single master node **(Batool & Byun, 2022)**. Further studies have also analyzed that smart contract transactions could reduce fraud risks helping improve predictability for 67.6% of freelancers who believe freelancing platforms provide better opportunities than traditional jobs **(Popereshnyak et al., 2024; Jabeen & Safdar, 2025)**.

The suggested platforms include "DLancer" which implements Ethereum blockchain and Smart Contracts to ensure issues regarding payment, then implementing Interplanetary File Storage (IPFS) for immutability **(Singh et al., 2024)**. Finally, advanced algorithms such as job recommender systems should be leveraged to pair freelancers with role based preferences **(Kunwar et al., 2024)**.

1.4 Key Takeaways from the Literature Review

With reference to the literature reviewed, the sector primarily operates through a "reputation economy", where trust is measured through scores. Although the sector itself has matured significantly, freelancers still experience cooperation gaps, i.e. 82% of freelancers express a desire for collaborative work, but current platforms are unable to satisfy their conditions. Furthermore, there is now a clear and necessary shift to a "distributed security responsibility" to protect the freelancers.

Comparisons between existing systems suggest that neither the traditional centralized models nor the fully decentralized suggested model is adequate for freelancers and clients requirements. While traditional frameworks offer superior scalability, lower latency, and multi-role workflows, they fail to resolve the high intermediary fees and data risks such as breaches and leaks. Similarly, pure decentralized systems are hindered by high transaction fees, and low user adoption.

Finally, to address weaknesses of both systems, literature points towards a hybrid Web3 architecture, which utilizes IPFS for decentralized storage, smart contracts and Distributed Ledger Technology (DLT) for transparency within the system, centralized Database for low latency searches and high-fidelity User Experience (UX) for user adoption.

1.5 Review of Similar Systems

To contextualize the project, some of the systems/freelancing web-applications reviewed were fiverr, upwork, WeWorkRemotely, Guru, PeoplePerHour and Ethlance.

1. **Traditional Centralized Hubs**

Currently established Web2 freelancing applications such as Upwork, Fiverr, Guru, Behance, WeWorkRemotely, and PeoplePerHour provide hourly or fixed contracts, employ milestone based payments, and rely on central servers and data centers. Other key features include reputation economy, reputation scoring, escrow based payments, etc.

While these platforms offer global reach and high comfort usability, the centralized system risks a single point of failure, which could lead to data risks, loss of data, and users themselves do not own their data. The lack of transparent financial logic or a public ledger deter freelancers from these systems.

2. **Web3 Prototype**

Ethlance, a decentralized Web3 freelancing application solved the issues regarding transparency, centralized data storage, and intermediary fees by operating on the Ethereum blockchain. However, as every interaction with the application required transaction with the Ethereum Mainenet, users were forced to pay “gas” fees for each and every action such as profile creation, profile updates, job posting, and other non-financial actions.

Furthermore, users' management of private keys, and crypto-wallets deterred user adoption to the platform resulting in the decline of web-applications in less than a year.

Platform	Model	Governance & Trust	Data Ownership & Storage	Primary Limitation
Behance	Centralized	Social validation No native escrow	Platform-controlled servers	No milestone enforcement
Guru	Centralized	Manual Arbitration SafePay Escrow	Centralized databases	Lower job volume
Upwork	Centralized	Centralized Monitoring Milestone Escrow	Platform-siloed history	High service fees Frequent scams
Fiverr	Centralized	Centralized Moderation Platform Payments	Centralized metadata & asset storage	Chargeback risks
WWR	Centralized	External Transactions Only	Simple job-board data center	Zero contract protection
PeoplePerHour	Centralized Hybrid	Manual Arbitration Escrow based deposits	Platform-controlled servers	High quality variance
Ethlance	Decentralized Prototype	Smart Contract Governance Smart Escrow	Fully on-chain (Mainnet)	Extreme gas costs Poor UX

Table 2: Review of Similar Systems

2. Project Purpose

2.1 Problem Statement

The rapid development of digital freelancing models has changed the global labor market, but the majority of the current models are designed with very centralized architectures posing a serious threat to structural and security risks. The centralized forms of governance put the absolute power to manage user information, reputation histories, and financial operations in the hands of the operators of these platforms, which is a single point of failure, data vulnerability, and does not give the freelancers real ownership. This involves high middlemen costs, late payments, non-transparent dispute resolution procedures, and single points of failure. (Shahzad, 2024)

Moreover, the existing Web2-based freelancing applications do not entirely cover these issues. Although blockchain technology makes it possible to create trustless payments, the current systems display a low level of integration between on-chain financial logic and off-chain content storage, which leads to the appearance of fragmented workflows, inefficient deliverable verification, and scaling challenges. The dependability of freelancers on the enforcement of escrows and dispute resolutions as well as identity verifications are undermining the concept of decentralization.

This creates a horrible need in the secure, hybrid decentralized freelancing protocol that guarantees ownership of the data, transparent enforcement of escrows, immutable validation of the deliverables, and user-controllable digital identity without compromising on performance or usability. (Kunwar et al., 2024)

This project will solve the issue of over-centralization, poor escrow, poor storage architecture and lack of sovereignty of the freelancers within existing freelance platforms by suggesting a decentralized but practical system design. It also points out the weakness of the centralized freelancing systems such as Upwork and Fiverr, which are the most popular in the gig economy but lead to problems such as high intermediary fees, late payments, unclear dispute resolution, and single points of failure that make their user data vulnerable to breaches and interruption.

2.2 Project Objectives

The core goal of the given project is to formulate and develop a hybrid decentralized freelancing protocol which advances safety, transparency, and autonomy of freelancers without impacting system efficiency.

The particular goals of the project are as follows:

- Implement a decentralized storage model with IPFS (through Kubo) to provide immutable and permanent storage of deliverables that address the issue of its loss through a central server and opens the opportunities to manage content in a censorship-resistant way.

- Create a logic layer using state-machine based smart contracts in Solidity to implement trustless escrow and financial mediation to solve payment delays and high fees in traditional platforms as well as creating transparent, deterministic transactions. (Singh et al., 2024)

- Create an identity layer based on the use of MetaMask and ECDSA signature recovery to enable secure authentication based on a wallet-oriented system with the identity and improving the lack of fraudulent transactions and the influence of identities over the users as a way to improve authenticity and mitigate the lack of trust. (Fulker and Riedl, 2024)

- Develop a hybrid design with off-chain metadata indexing (PostgreSQL) and on-chain security anchors to provide a balance between performance and decentralization to reduce the scalability problems inherent to pure blockchain systems and to allow useful job matching and searches.

- Implement frontend/backend (JS/HTML, Ethers.js, and FastAPI) to facilitate user interactions and address the usability limitations of the initial prototypes of the Web3, as well as to provide the possibilities of natural bidding, hiring, and collaboration in a decentralized setting. (Kunwar et al., 2024)

- To offer a platform on which the freelancer ecosystem can be fairer, by decreasing middleman power, middleman charges, and unequal influence, and thus allowing Freelancers to save their work history, reputation, and earnings. (Singh et al., 2024).

No.	Proposed Functionality	Problems Solved / Opportunities Created
1	Implement decentralized storage layer using IPFS (via Kubo) for immutable deliverables	Solves data loss, breaches, and central server downtime; creates censorship-resistant, permanent content storage

No.	Proposed Functionality	Problems Solved / Opportunities Created
2	Develop state-machine-based smart contracts in Solidity to trustless escrow and payments	Removes high intermediary fees (10-20%), and delayed payments (5-14 days).
3	Create wallet-based identity layer with MetaMask and ECDSA signature recovery	Solves fraud, pseudo-identity creation and loss of user control; offers secure and tamper-resistant, passwordless authentication and ownership of the user. (Shahzad, 2024; Singh et al., 2024)
4	Create Hybrid architecture, off-chain metadata indexing (PostgreSQL) + on-chain anchors	Balances performance and decentralization; mitigates scalability and gas fee issues in pure blockchain systems while enabling fast searches and lookups (Kunwar et al., 2024).
5	Hybrid & distributed project matching and structured search via backend services (FastAPI)	Enhances the efficiency of freelancer-client matching and workforce optimalization, which has been observed to be inefficient in current gig economy platforms. (Kunwar et al., 2024)
6	Compare protocol based on simulations and testing (latency less than 1.5s, storage availability more than 95 percent)	Closes the research gaps in empirical validation of decentralized freelancing; provides quantifiable security, reliability, and practicality.
7	Cryptographically linked deliverables and escrow state transitions	Closes the disconnect between financial settlement and content verification by linking submitted work (CID) with contract state, enhancing agreement enforcement (Singh et al., 2024).
8	Transparent and publicly verifiable contract state transitions	Increases fairness and accountability, as all parties are able to audit the contract execution without depending on centralized arbitration authorities (Gandini et al., 2016).

Table 3: Project Objectives

3. Project Description, Scope and Management Milestones

3.1 Project Description

This project aims to address the limitations of traditional centralized freelancing platforms by designing a decentralized, web based freelancing platform that makes trust and improves transparency and collaboration between clients and freelancers. To address challenges that are found in traditional freelancing platforms such as unreliable identities, opaque workflows and inefficient talent discovery this system will be using blockchain based identity verification, rule based matching and structured data management.

This system introduces MetaMask which will be used for identity management allowing users to authenticate and users will also interact with the platform through blockchain based identities. In this system the users will use cryptographic wallets to establish identity instead of traditional username and password authentication and by using this method we will reduce reliance on centralized credentials storage hence improving the overall security.

One of the core components of the platform will be the use of deterministic smart contracts developed in solidity which will be used to manage project agreements and transaction verification workflows. The smart contracts will define the milestones, completion status and condition of the project and enforce platform rules related to project agreements and ensure tamper-resistant execution of core logic.

To support decentralized data handling, project related assets such as documents will be hosted using distributed storage technology such as Kubo/IPFS, which will reduce dependency on centralized servers. A rule based matching algorithm will be implemented using FastAPI, where the projects and freelancers are matched based on predefined logical rules such as skills, roles, availability and project requirements. This platform will also add structured search and data operations using postgreSQL which will be used to manage user profiles, their project listings.

Overall, this project demonstrates how to integrate rule based logic, distributed storage, smart contracts and decentralized identity into a modern freelance platform. The system remains practical while effectively demonstrating distributed system and blockchain principle by focusing on fundamental architectural ideas and excluding production level features like financial gateways and complex collaboration tools.

3.1.1 Proposed architecture and system workflow

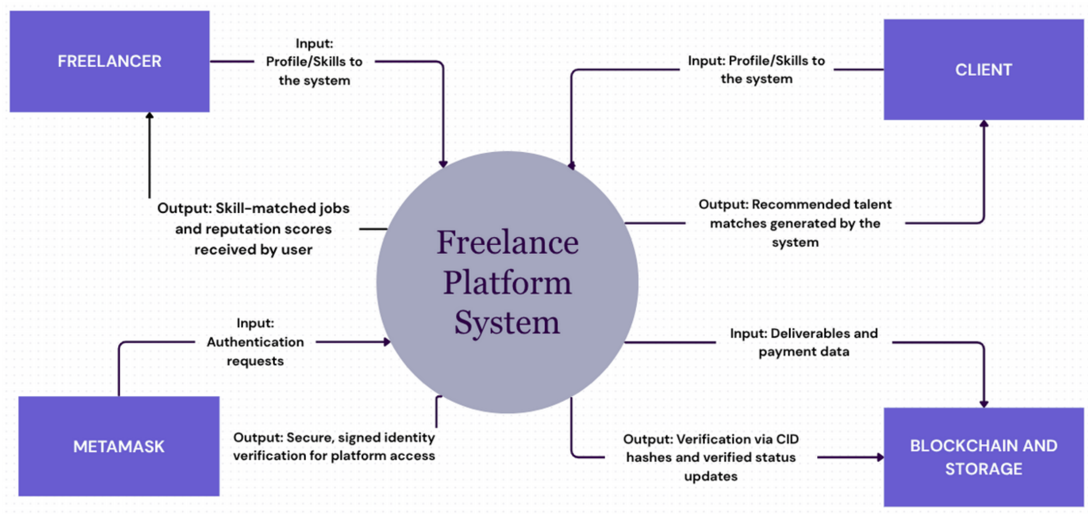

Fig.1 - System Context Diagram of the Freelancing Platform

Figure 1. shows how external entities and the core system interact as well as the proposed freelancing platform's overall system context. Connecting clients, freelancers, wallet based identity services, decentralized storage, and blockchain components, the platform serves as a central coordination layer. The platform generates skill matched recommendations and project related results using a rule based matching mechanism while freelancers and client supply the system with profile and skill related information.

MetaMask handles authentication requests, allowing users to authenticate themselves using cryptographic wallet signatures rather than conventional login credentials. Blockchain services and decentralized storage are used to store project deliverables and verification metadata. These services provide the system with verified status updates and content identifiers (CIDs). This architecture exemplifies a distributed design in which separate components interacting via network based protocols handle identity, storage and verification tasks.

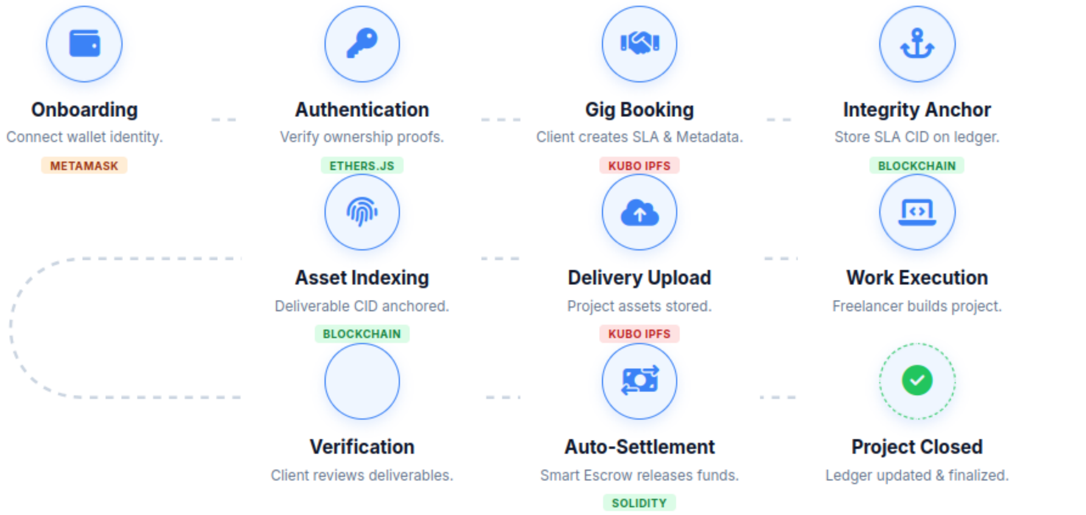

Fig.2 - System Workflow Diagram

Figure 2. Represents the entire process of the proposed freelance platform, from project closure to user onboarding. Wallet based onboarding is the first step in the process, during which users link their MetaMask wallet and use cryptographic signatures to verify ownership. Clients create project agreements and metadata after being authenticated and Kubo IPFS is used to store them in decentralized storage.

Deterministic smart contracts are used to secure the agreements content identifier (CID) on the blockchain, guaranteeing integrity. The deliverable CID is then once more recorded on the blockchain for verification purposes after freelancers complete the assigned work and upload deliverables to IPFS. the smart contract updates the project status and completes the transaction workflow after clients review and approve the submitted work without incorporating actual financial settlement systems, this procedure guarantees transparency, immutability and verifiable execution.

3.2 Scope

The system will use MetaMask to implement wallet based identity management allowing users to interact with the platform and authenticate using identities backed by blockchain technology rather than the traditional username password method

To manage project workflows, agreements and transaction verification logic, the project will include deterministic smart contracts created in solidity. Instead of managing actual financial settlements, these smart contracts will enforce predetermined rules that relates to the project states and milestones, guaranteeing transparency and consistency in execution.

Project related digital assets including deliverables and documents will be stored using Kubo (IPFS) for decentralized data handling allowing for distributed and tamer resistant storage. In order to facilitate dependable and effective search and retrieval processes, postgreSQL will be used to manage structured platform data such as user profiles project listing and metadata.

FastAPI will be used to implement a rule based matching algorithm that will suggest qualified freelancers for projects based on predetermined parameters like roles skills and project requirements.

3.2.1 Out of scope features

Mainnet deployment and smart contract gas optimization are not included in the project, blockchain functionality will only be used in test or local network environments for demonstration. Since the project focuses on transaction logic and verification rather than financial processing, currency settlement, banking systems and payment gateway integrations are completely excluded.

Advanced collaboration features, such as real time communication or video conferencing via VoIP are out of scope. Also, DAO-based decentralized mechanisms for dispute arbitration will not be implemented. The system shall be developed only as a web-based application, cross platform native mobile applications are beyond the current scope.

This project integrates decentralized identity, smart contracts, distributed storage, rule based matching, and structured data management within the scope of a controlled and achievable academic scope. By clearly defining the functional boundary and excluding production level features, it effectively shows the concepts of modern distributed systems and blockchain.

3.3 Management milestones

Milestone No.	Milestone	Responsible	Expected Duration (days)
1	Project Ideation	All Members	3
2	Feasibility Analysis	All Members	4
3	Literature Review	All Members	10
4	Requirement Gathering and Analysis	All Members	6
5	System Design & Architecture	Sarun Maharjan, Runa Maphu	9
6	Resource Allocation & Tool Setup	Sarun Maharjan	4
7	Proposal Documentation	All Members	5
8	Smart Contract Design & Development	Sarun Maharjan	10
9	Database design	Pawan Poudel	12
10	Rule-Based Matching Logic Development	Sarun Maharjan, Anushree Pradhan	10
11	User Interface wireFrame & Design	All Members	11
12	Frontend Development	Bijee Dangol	20
13	Backend Development & API integration	Anushree Pradhan	20
14	IPFS (KUBO) integration	Pawan Poudel	10
15	System Integration	Sarun Maharjan, Runa Maphu	15
16	Unit Testing	All Members	8
17	Integration Testing & System Testing	All Members	9

18	Documentation, Evaluation & Final Submission	All Members	7

Table 4: Management Milestones

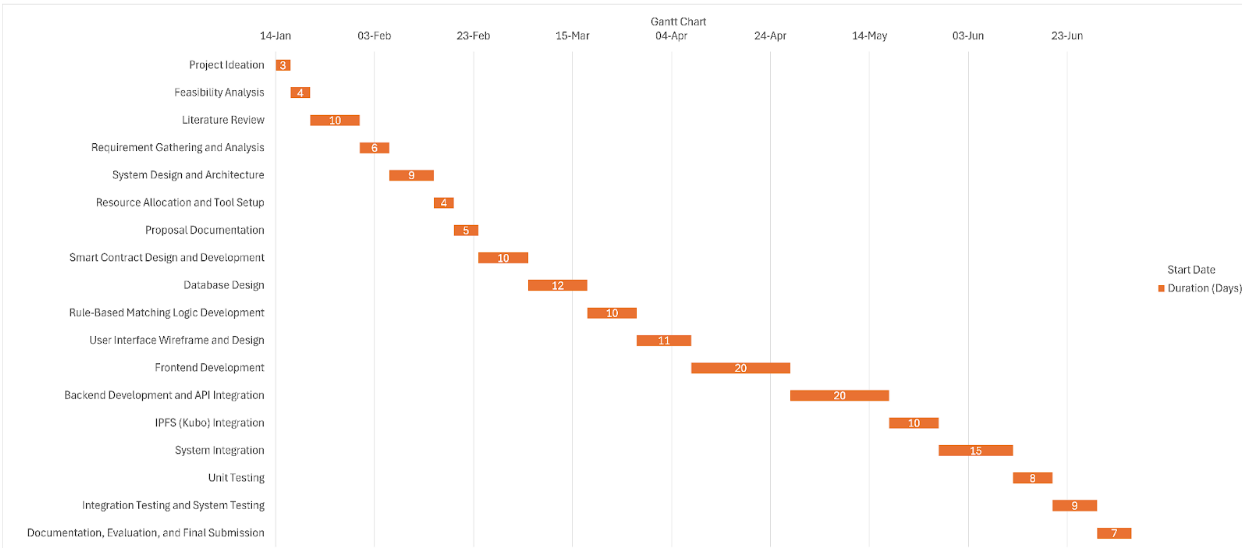

Fig. 3 - Gantt Chart for Project Milestones

4. Project Organization

4.1 Organization Description

The project team, designated as Group 19, was formed to address the structural and security limitations of centralized freelancing platforms. The team is composed of four members, each bringing specialized skills to the development of a decentralized Web3 protocol.

The project will be executed in the form of a role-based team: the development of the frontend, the implementation of backend services, integration of the blockchain, management of the database, and the development of the workflow of the system will be carried out by specific members based on their technical skills. Sarun Maharjan, the Project Manager and development head, manages to allocate tasks, coordinate progress and integrate the system components.Everyone from this group is responsible for research and also for documentations of their parts which are segregated by our documentation and development lead; Sarun Maharjan.

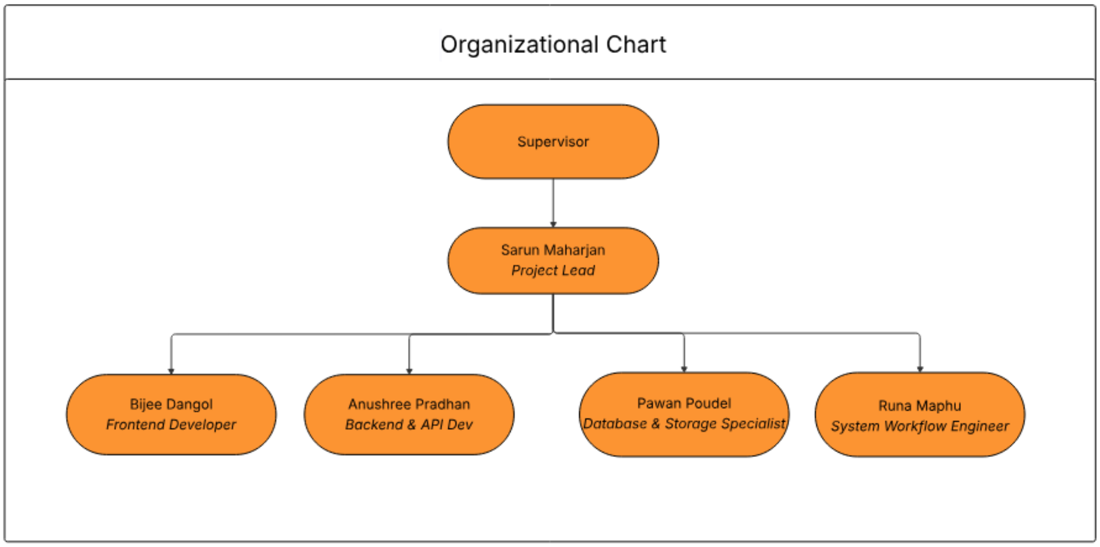

Fig. 4 - Organisational chart

4.2 Roles and Responsibilities

The following table details the specific roles and responsibilities of each stakeholder involved in the project.

Full Name	Roles	Responsibilities
Project Supervisor	Academic Supervisor	He/she provides academic oversight, reviews milestones, and ensures the project meets university standards.
Sarun Maharjan	Project Manager / Backend & Blockchain Developer	He coordinates team activities, manages project milestones, oversees system integration and designs smart contracts, implements MetaMask identity layers, and develops matching algorithms.
Anushree Pradhan	Backend & API Developer	She designs and implements backend services using FastAPI, develops RESTful APIs, handles server side

		business logic, manages authentication workflows and integrates backend services with blockchain and database components.
Bijee Dangol	Frontend Developer	She develops the web interface, integrates blockchain interaction layers, and manages user experience design.
Pawan Poudel	Backend Database & Storage Specialist	He manages PostgreSQL indexing, implements IPFS storage, and handles off-chain metadata.
Runa Maphu	System Workflow Engineer	She designs and implements project workflows, role-based interactions, and state transitions within the application logic.

Table 5: Roles and Responsibilities

5. Risk Analysis

This section identifies potential risks to the project, assesses their likelihood and impact, and outlines mitigation strategies. A Severity Matrix is used to categorize the overall risk level.

This section identifies and analyzes the potential risks associated with the project, assesses their likelihood and impact, and outlines mitigation strategies. To categorize the overall risk levels; a severity matrix is used and the following table is the severity matrix:

5.1 Potential Risks and Mitigation Strategies

Potential Risk	Likelihood	Impact	Mitigation Strategy
Smart Contract Vulnerabilities	Medium	High	A rigorous unit testing and simulations of Solidity contracts should be conducted.

IPFS Data Availability Issues	Medium	Medium	A hybrid storage model with off-chain metadata indexing in PostgreSQL should be implemented. IPFS pinning services should be utilities to ensure data persistence and availability.
Low User Adoption of Web3 Tools	Medium	High	A user friendly onboarding documentation and tutorials should be provided. Also, an intuitive frontend should be designed using Ethers.js to abstract away complex blockchain interactions and should focus on a seamless UX.
Human Uncertainty & Trust in Decentralized Systems	High	Medium	A robust state machine based escrow system that requires mutual validation or milestone based releases should be implemented.
Privacy Breaches & Data Leakage from Off-chain Components	Medium	High	A “distributed security responsibility" model should be adopted. And all off-chain data such as in PostgreSQL is encrypted at rest and in transit should be ensured.
Integration Complexity	High	Medium	Modular architecture and continuous integration testing components must be used.
Lack of Skilled Developers for Specific Web3 Technologies	Medium	Medium	Should leverage existing team members' diverse and specific skill sets and should allocate time for upskilling in the allocated areas(eg. Solidity development, IPFS Integration) by utilizing open source resources and communities.
Regulatory Uncertainty in Web3 Space	Low	High	Should monitor evolving blockchain regulations. Also, focus on academic demonstration.

Table 6: Potential Risks and Mitigation Strategy

6. Appendix

6.1 Literature Matrix Table

Works	**Data Source**	**Context**	**Objectives**
(Shahzad, 2024)	Luleå University of Technology	Information Security	Explore platform vulnerabilities, implementing TAM/ IPT theories.
(Kunwar et al., 2024)	IARJSET	Job Matching Systems	Explore algorithmic job matching through listed skills and experience.
(Popereshnyak et al., 2024)	CEUR Workshop	Blockchain Platforms	Implement Ethereum smart contract to develop a secure freelancing platform.
(Bari et al., 2015)	Beihang Univ.	Industrial Practices	Summarize lessons from crowdsourcing and identify future research.
(Alvarez de la Vega et al., 2023)	ACM CSCW	Work-Life Balance	Work-life experience of freelancers through qualitative study.
(Batool & Byun, 2022)	Electronics (MDPI)	Fraud Prevention	Integrate blockchain to reduce fraud in freelancing platforms.
(Blaising & Dabbish, 2022)	ACM CSCW	Worker Socialization	Identify challenges faced by freelancers when initially starting freelancing.
(Fulker & Riedl, 2024)	ACM CSCW	Gig Cooperation	Examine freelancers willingness to work collaboratively compared to individually.
(Gandini et al., 2016)	Middlesex Univ. / Elance Data	Reputation Economy	Analyze how reputation scores correlate between earnings and act as a proxy for trust.

(Gupta et al., 2020)	Processes (MDPI)	Software Engineering	Map research on how freelancers catalyze software development and the decision factors for companies/freelancers.
(Jabeen & Safdar, 2025)	Qlantic Journal of Social Sciences	Job Satisfaction	Understand the impact of freelancing platforms on professional development, and impact on digital workers.
(Rauf et al., 2023)	arXiv / Open University	Security Responsibility	Introduction to distributed security responsibility in freelance software development.
(Singh et al., 2024)	Nitte Meenakshi Inst. Tech	Decentralized Systems	Build “DLancer”, a decentralized platform through smart contracts, and DAO.
(Tamsetwar et al., 2026)	IRJMETS	Web3 Architecture	Build “Delance”, a hybrid Web3 architecture to address high fees and opaque transactions.
(Thant & Htet, 2025)	Univ. of Computer Studies	Marketplace Systems	Develop a Django-based marketplace for freelancers with role-based management.

6.2 Research Process Diagram